MILITARY SPECIFICATION

CHAFF, COUNTERMEASURES RR-129/ALE-29A

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The Radar Reflector Package covered by this specification shall consist of a plastic case loaded with aluminum-coated glass chaff. The package shall be dispensed from an AN/ALE-29A Chaff Dispenser Set to produce confusing signals in radar systems operating in the frequency range of 2,500 - 10,500 MHz.

2. APPLICABLE DOCUMENTS

*2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-F-320	Corrugated and Solid, Sheet Stock (Container Grade) and cut shape
PPP-B-640	Boxes, Fiberboard, Corrugated Triple Wall
PPP-T-60	Tape: Pressure-Sensitive Adhesive Waterproof, for Packaging

Military

MIL-B-131	Barrier Material: Water Vaporproof, Flexible
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-T-18303	Test Procedures: Preproduction and Acceptance for Aircraft Electronic Equipment, Format for

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MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-129	Marking for shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

DRAWINGS

Military

NWL DWG. 2519577	Naval Weapons Laboratory, Dahlgren, Source Control Drawing for Cartridge, Impulse

Lundy Electronics & Systems, Inc.

LDY DWG. 39-0030	Case, Assembly

NAVAL AIR SYSTEMS COMMAND

WR-43	Preparation of Quality Assurance Provisions (Bureau of Naval Weapons)

2.1.1 Availability of Documents

(1) When requesting specifications, standards and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, CODE 105, 5801 Tabor Avenue, Philadelphia, PA 19120. Drawings must be ordered from the activity which originated the drawing or the Contracting Office.

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration.

3.2.1 Selection of Materials - Specifications and Standards for all materials, parts and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

*3.2.2 Chaff Material - The chaff material shall be a Type E glass monofilament coated continuously with 99.0% or purer aluminum. The aluminum coating thickness shall be 0.12 ± 0.06 mils with a glass substrate diameter sufficient to produce a coated fiber diameter (reference) of 1 mil.

*3.2.2.1 Acceptable Dipoles - A minimum of 90% of the dipoles in each chaff round shall be acceptable. An acceptable dipole shall be one whose DC resistivity measures less than 10 ohms per linear inch and whose metal coating is continuous along its length and around its circumference. One hundred dipoles from each chaff round selected for sampling tests, shall be examined for compliance with these requirements by means of repeatable optical and electrical measurements.

*3.2.3 Secondary Coating - A slip coating shall be applied as a secondary coating to the chaff material to: (1) prevent dipole end welding and metal smear during the cutting operation; (2) prevent cohesion between dipoles which may be caused by corrosive conditions; and (3) increase dispersion by reducing dipole-to-dipole friction. Each chaff unit selected for testing, shall retain 75% of the dispersibility of an unexposed bundle after being subjected to humidity test, specified in MIL-T-5422, when tested in a calibrated low velocity vertical wind tunnel.

3.3 Design and Construction - The chaff rounds shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Standard Conditions - The conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

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Temperature	Room ambient (25°C± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity

3.3.2 Service Conditions - The chaff rounds shall operate satisfactory under any of the environmental service conditions or reasonable combinations of these conditions as specified in Specification MIL-E-5400 for Class 4 equipment, except as modified herein.

3.3.2.1 Vibration - The chaff rounds shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400.

3.3.2.2 Altitude - 50,000 feet

3.3.2.3 Shock - 20 g.

*3.3.2.4 Temperature - Continuous requirements only.

*3.3.3 Identification of Product - Nomenclature assignment shall be in accordance with MIL-N-18307. Identification shall conform to MIL-STD-130. Rounds shall be identified by embossed white lettering against a medium blue background on the end cap as follows:

RR-129/ALE-29A, E, G, and I-Band Chaff
FSN 5895-929-6095-NW20

3.4 Performance - The chaff round to be supplied hereunder shall be suitable for dispensing from an AN/ALE-29A Chaff Dispensing Set. The chaff shall produce a chaff echo in radar systems within the frequency range of 2,500 - 10,500 MHz. The individual chaff round shall simulate an airborne vehicle whose mean value of radar cross-section is a minimum of: 100 square meters for rounds falling within the frequency range of 2,500 - 3,500 MHz; 75 square meters for rounds falling within the frequency range of 4,500 - 6,500 MHz; and 50 square meters for rounds falling within the frequency range of 8,300 - 10,500 MHz. The chaff is required to attain these values of cross-section two seconds after ejection in at least eight out of ten trials at the resonant frequency within each of the aforementioned frequency bands.

3.5 Detail Requirements

3.5.1 Chaff - The chaff unit supplied hereunder shall meet the following requirements.

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*3.5.1.1 Form Factor - The dipole lengths, chaff unit weights and the number of chaff units required per shaff round are shown in Table I.

*3.5.1.1.1 Packaging Density - Each chaff unit shall have a circular cross-section. The weight density of each chaff unit shall be 23.0 ± 2.3 gms/in^3 when packaged in the case assembly, to achieve an optimum dipole packing density of 45% ± 4.5% of a theoretical hexagonal array. The weight density of each chaff unit in a chaff round shall be determined to insure compliance with the requirements of Table I in accordance with the provisions of paragraph 4.4.2.

*3.5.1.2 Assembly - The chaff units shall be separated by 11 point beaming board dividers. The cumulative tolerance of all chaff units in any one round shall not exceed a positive 0.070 inch. The chaff units shall be positioned within the chaff case in accordance with LDY Drawing 39-0030.

3.5.2 Case Assembly

3.5.2.1 Function - The case assembly shall be completely compatible with and suitable for dispensing of the chaff payload from an AN/ALE-29A dispenser by the impulse cartridge defined by the U. S. Naval Weapons Laboratory Source Control Drawing 2519677. The case assembly shall provide a container for the chaff payload capable of normal handling including repeated insertion and removal from an AN/ALE-29A dispenser. The assembly shall retain all components except the chaff payload, piston and end cap when used with the above dispenser and cartridge. The case assembly shall also be designed such that its use with the above cartridge shall preclude any damage, scoring or charring of the AN/ALE-29A dispenser, or premature release of chaff under the service conditions specified in paragraph 3.3.2. The case assembly shall be so designed as to permit easy assembly or disassembly, by hand, or cartridge and case.

3.5.2.2 Form Factor - The form factor for the case assembly shall be compatible with its intended use as specified in paragraph 3.5.2.1 above and shall be subject to the approval of the procuring activity.

*3.5.2.3 Contents - The case assembly shall consist of the components shown in LDY Drawing 39-0030.

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*TABLE I

Chaff Units Per Round	Chaff Unit Length (Inches)	Chaff Unit Weight (Grams)	Number of Dipoles per Chaff Unit
1	2.00 ± 0.020	67.0 ± 6.7	750,000 ± 10%
1	1.12 ± 0.020	37.0 ± 3.7	750,000 ± 10%
3	0.61 ± 0.020	20.0 ± 2.0	750,000 ± 10%

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests

*4.1.2 Classification of Characteristics - The characteristics verified by the tests and examinations herein are classified as major or minor in accordance with WR-43. Tests and examinations that verify major characteristics are identified by the symbol (M). Tests and examinations which are not annotated with a classification code are classified minor.

*4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on chaff rounds representative of the production units to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of paragraph 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the chaff rounds meet all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and rough handling tests in accordance with the requirements in Appendix C of MIL-STD-794.

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4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production chaff rounds shall be delivered prior to the approval of the preproduction sample. Prefabrication of production chaff rounds prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of chaff rounds to be submitted for acceptance. The preproduction sample shall not be considered as chaff rounds under the contract.

4.2.4 Production Chaff Rounds - Chaff rounds supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each chaff round shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and, for units already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring agency.

4.3 Initial Production Tests - One hundred twenty of the first one thousand chaff rounds shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. These chaff rounds shall be selected by the procuring activity after the chaff rounds have successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - These chaff rounds may be subjected to any and all tests the procuring activity deems necessary to assure that the production chaff rounds are equivalent to the previously approved preproduction sample in design construction workmanship, performance and quality, and that they meet all applicable requirements.

4.3.2 Accessory Material - In addition to the chaff rounds submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the chaff rounds.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Procution Sample to pass any of the tests shall be the cause for deliveries under the contract to cease until proper corrective action is approved and accomplished. Corrective action

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shall also be accomplished on chaff rounds previously accepted when requested by the procuring activity.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Special Tests

*4.4.1 Individual Tests (M) - Each chaff round submitted for acceptance may be subjected to an individual test. This test, an examination of product, shall be adequate to determine compliance with the requirements of material, workmanship and operational adequacy.

4.4.2 Sampling Tests - Chaff rounds selected for sampling tests shall first have passed the individual tests. One out of every one thousand chaff rounds shall be selected for sampling tests by the Government Inspector.

*4.4.2.1 Scope of Tests -

(1) Sampling tests shall include all tests deemed necessary by the procuring activity to determine that the chaff rounds meet all the requirements of this specification and the contract. Sampling tests shall include optical and electrical measurements to insure compliance of the product with the requirements specified in paragraph 3.2.2, and environmental tests in accordance with the procedures of Specification MIL-T-5422, simulating the service conditions specified in paragraph 3.3.2.

(2) Sampling tests of chaff rounds shall also include operational tests to be conducted following environmental tests of these same chaff rounds simulating the service conditions specified in paragraph 3.3.2. The Operational Tests shall include live firings from an AN/ALE-29A Chaff Dispenser to

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demonstrate compliance with the functional requirements of paragraph 3.5.2.1 and overall effectiveness of the payload including satisfactory discharge and separation of the dipoles, retention of the case assembly (except the cap and piston) and ease of removal of the case assembly after firing.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of chaff rounds for the purpose of checking the effect of any design or material change on the performance of the chaff rounds and to assure adequate quality control. The chaff rounds selected for special tests may be selected from chaff rounds previously subjected to the sampling tests.

4.4.3.1 Special Test Schedule - Selection of chaff rounds for special tests shall be made as follows:

(1) On early chaff rounds after an engineering or major change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as required by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Chaff Round Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.
(2) Determine if the failure is an isolated case or design defect.
(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide

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for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

5. PREPARATION FOR DELIVERY

5.1 Packaging of Rounds in Inner Carton - Thirty rounds shall be packaged per carton. Each round shall be separated by a laminated board filler or egg crating conforming to PPP-F-320. The filler material shall not be crushed or deformed in handling, storing, or in the conduct of the test specified in paragraph 5.2.1. The filler shall also be designed to support the chaff round such that it is not damaged during normal handling or in the conduct of the tests specified in paragrpah 5.2.1.

*5.1.1 Carton Details - The unit container shall be style RSC (regular slotted box) conforming to PPP-B-636, Type CF, class-domestic, Variety SW, Grade 200. A closure tape conforming to PPP-T-60 shall seal the exposed edges of the box.

*5.1.2 Barrier Bag - The inner carton containing the individual rounds of chaff shall be enclosed in a close-fitting heat-sealed bag, fabricated from barrier material conforming to MIL-B-131, Class 1. The air volume within the barrier material whall be reduced to a practicable minimum before sealing the barrier.

5.2 Packing - Eight cartons shall be packed in an exterior fiberboard package for overseas shipment in accordance with PPP-B-640, Class 2, Grade A, RSC. The exterior dimensions of the outer package shall not exceed two feet in width, depth and height.

5.2.1 Preproduction Tests - Preproduction tests of a loaded exterior fiberboard package shall be performed in accordance with the requirements in Appendix C of MIL-STD-794, except that preproduction chaff rounds shall be used in lieu of dummy loads.

*5.3 Marking of Shipments - Interior cartons and exterior packaging shall be marked in accordance with MIL-STD-129 and include the legend "CHAFF, COUNTERMEASURES RR-129/ALE-29A for E-, G-, I-band, FSN 5895-929-6095-NW20.

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6. NOTES

6.1 Ordering Data - Procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable levels of packaging and packing (see paragraph 5.1).

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Revisions - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81467B(AS) CHAFF, COUNTERMEASURES RR-129/ALE-29A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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